

Outlines

- Strength/weakness of DES, AES
- Public Key Cryptography
- Modular Arithmetic
- RSA

Avalanche Effect

- Key desirable property of encryption alg
- Where a change of one input or key bit results in changing more than half output bits
- DES exhibits strong avalanche
- Even flipping one input bit typically changes about 32 out of 64 output bits
- This strong diffusion is achieved through:
 - 16 Feistel rounds
 - S-boxes and P-box permutations

Strength of DES - Key Size

- 56-bit keys have $2^{56} = 7.2 \times 10^{16}$ values
- Brute force search appeared impractical
- Recent advances have shown is possible
 - in 1997 on a huge cluster of computers over the Internet in a few months
 - in 1998 on dedicated hardware called "DES cracker" by EFF in a few days (\$220,000)
 - in 1999 above combined in 22hrs!
- Even after cracking, Still must be able to recognize plaintext
- No big flaw for DES algorithms, real weakness is short 56-bit key

DES Replacement

- Triple-DES (3DES)
 - 168-bit key, brute force attacks impossible
 - Underlying encryption algorithm the same, no effective analytic attacks
 - Drawbacks
 - Performance: no efficient software codes for DES/3DES
 - Efficiency/security: bigger block size desirable
- Advanced Encryption Standards (AES)
 - US NIST issued call for ciphers in 1997
 - Rijndael was selected as the AES in Oct-2000

AES

- Private key symmetric block cipher
- 128-bit data, 128/192/256-bit keys
- Stronger & faster than Triple-DES
- Provide full specification & design details
- Evaluation criteria
 - Security: effort to practically cryptanalysis
 - Cost: computational efficiency and memory requirement
 - Algorithm & implementation characteristics: flexibility to apps, hardware/software suitability, simplicity

AES Shortlist

- After testing and evaluation, shortlist in Aug-99:
 - MARS (IBM) - complex, fast, high security margin
 - RC6 (USA) - v. simple, v. fast, low security margin
 - Rijndael (Belgium) - clean, fast, good security margin
 - Serpent (Euro) - slow, clean, v. high security margin
 - Twofish (USA) - complex, v. fast, high security margin
- Then subject to further analysis & comment

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Private-Key Cryptography

- Private/secret/single key cryptography uses one key
- Shared by both sender and receiver
- If this key is disclosed communications are compromised
- Also is **symmetric**, parties are equal
- Hence does not protect sender from receiver forging a message & claiming is sent by sender (non-Repudiation)

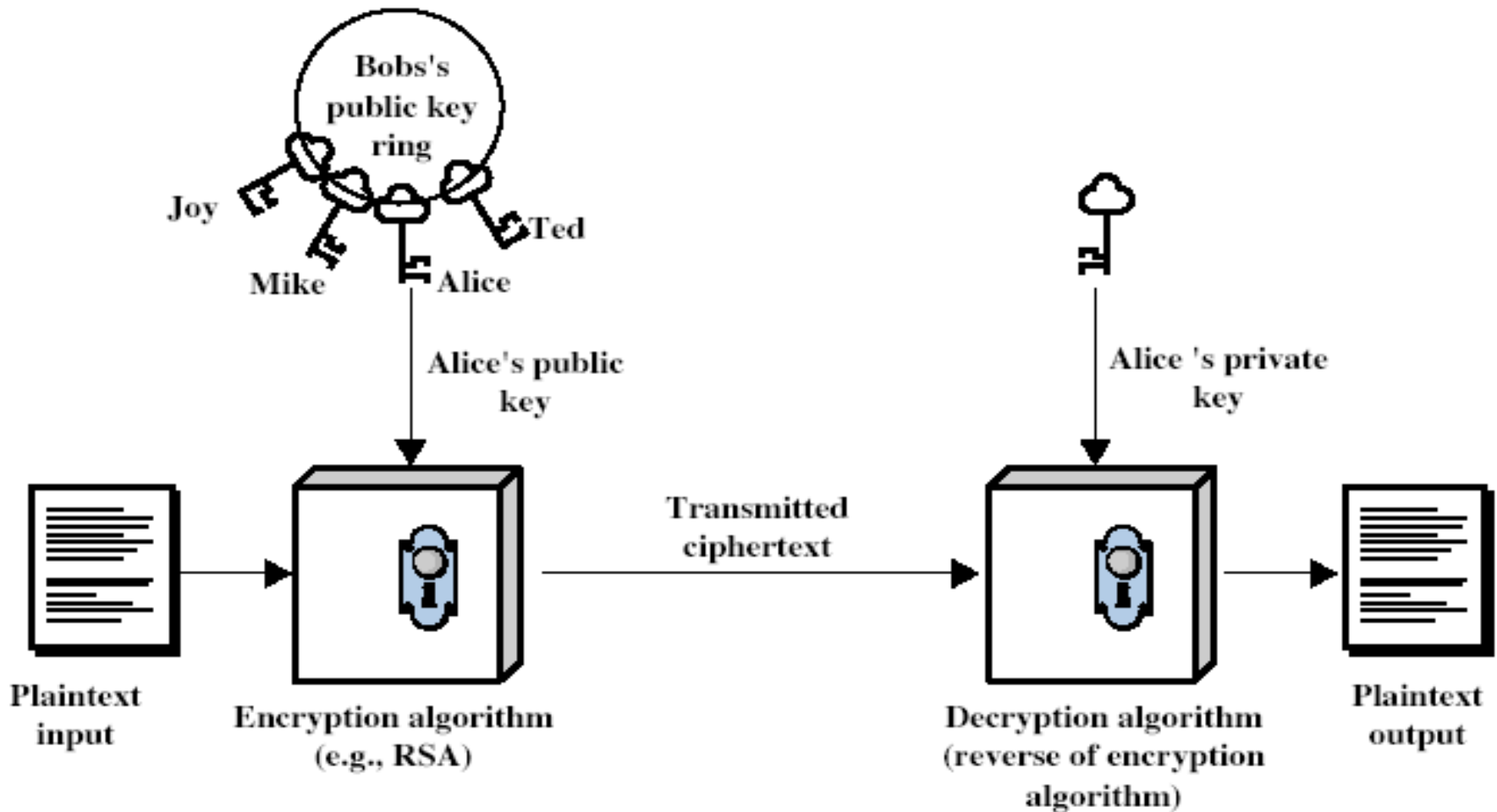
Public-Key Cryptography

- Probably most significant advance in the 3000 year history of cryptography
- Uses two keys - a public & a private key
- **Asymmetric** since parties are not equal
- Uses clever application of number theoretic concepts to function
- Complements rather than replaces private key crypto

Public-Key Cryptography

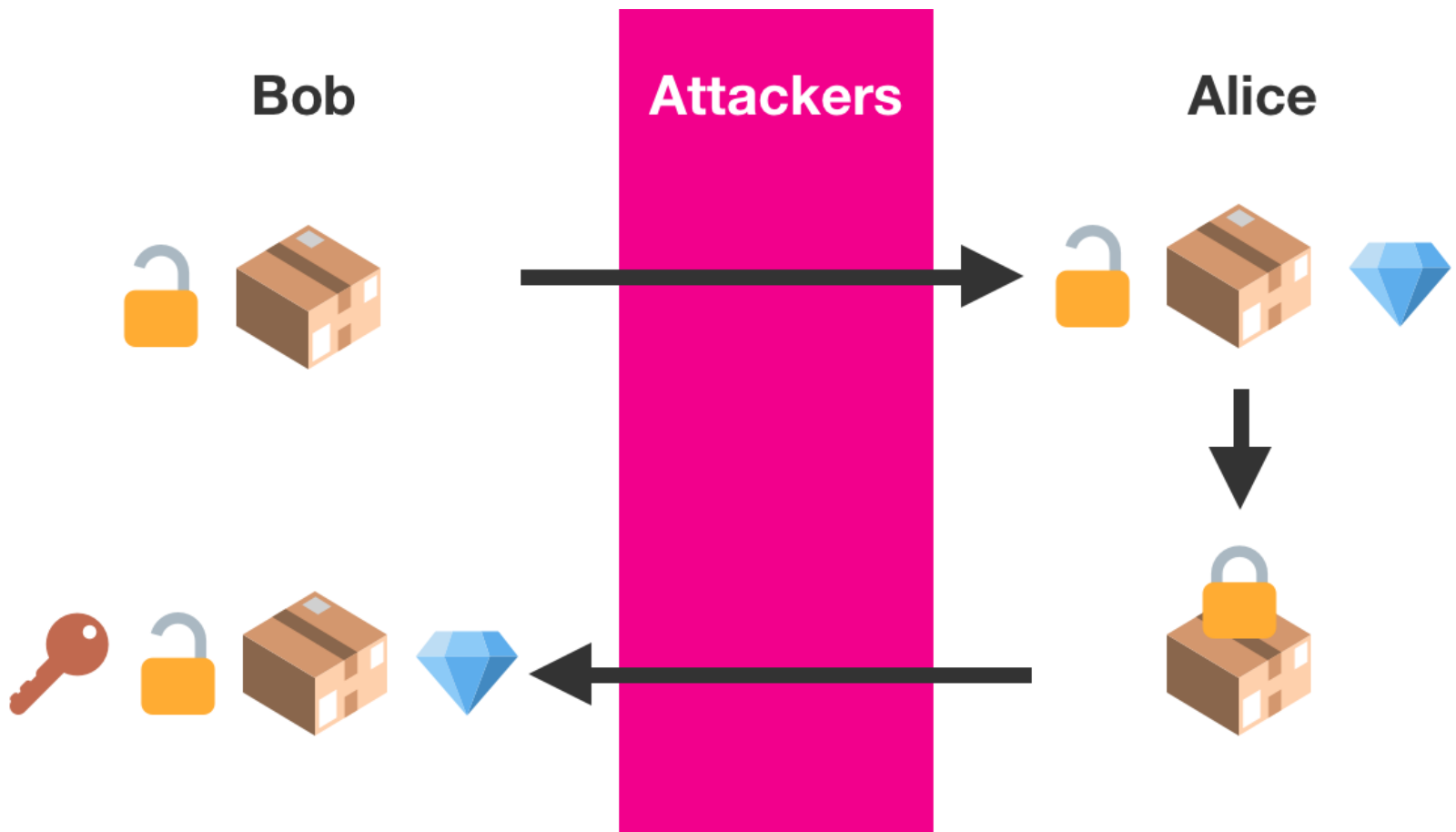
- Public-key/two-key/asymmetric cryptography involves the use of two keys:
 - a public-key, which may be known by anybody, and can be used to encrypt messages, and verify signatures
 - a private-key, known only to the recipient, used to decrypt messages, and sign (create) signatures
- Asymmetric because
 - those who encrypt messages or verify signatures cannot decrypt messages or create signatures

Public-Key Cryptography



Public-Key Characteristics

- Public-Key algorithms rely on two keys with the characteristics that it is:
 - computationally infeasible to find decryption key knowing only algorithm & encryption key
 - computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known
 - either of the two related keys can be used for encryption, with the other used for decryption (in some schemes)
- Analogy to delivery w/ a padlocked box



Public-Key Cryptosystems

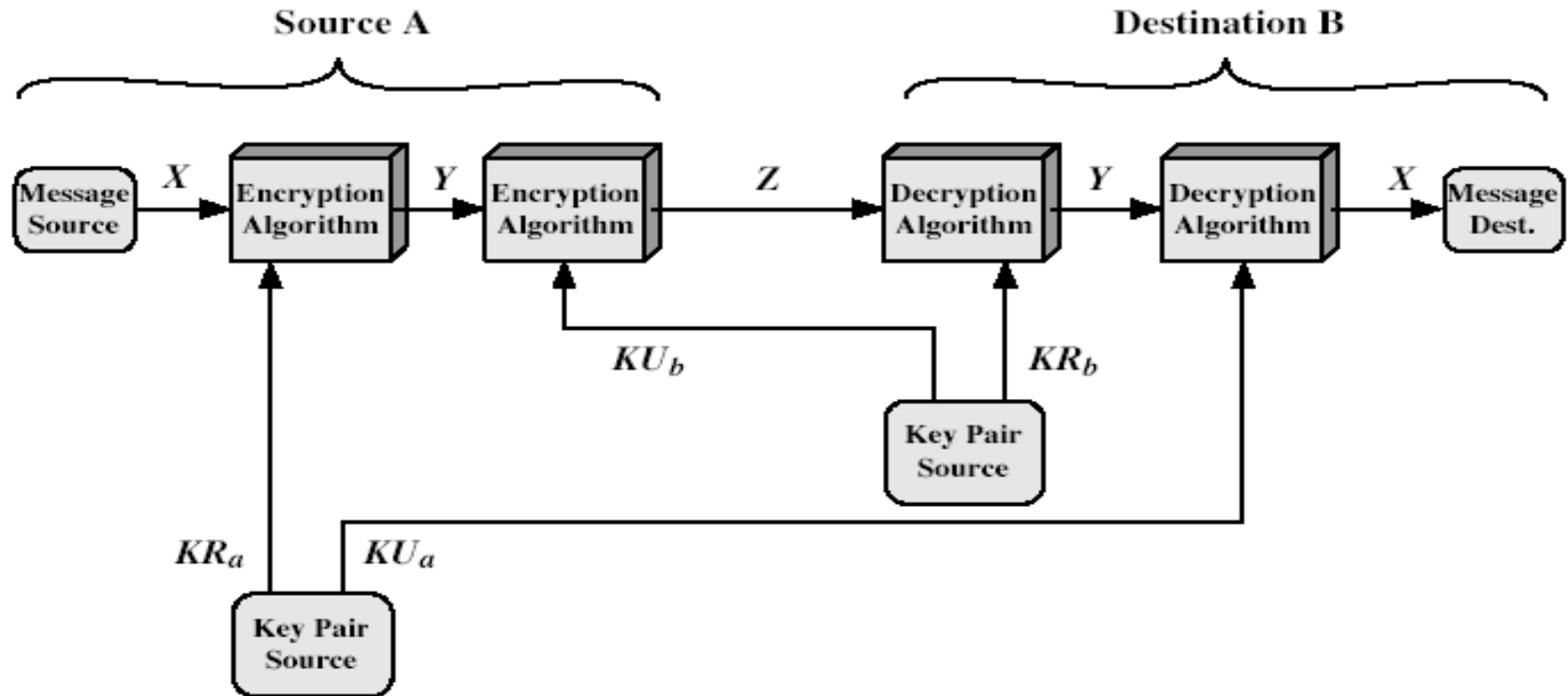


Figure 9.4 Public-Key Cryptosystem: Secrecy and Authentication

- Two major applications:
 - encryption/decryption (provide secrecy)
 - digital signatures (provide authentication)

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Modular Arithmetic

- Public key algorithms are based on modular arithmetic.
- Modular addition.
- Modular multiplication.
- Modular exponentiation.

Congruence

If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, then $a \equiv b \pmod{m}$ if and only if $m \mid (a - b)$

Modular Addition

Rows shows d_m , and each column is the key digit d_k

- Addition modulo (mod) K

- Poor cipher with $(d_k + d_m) \bmod K$, e.g., if $K=10$ and d_k is the key.

+	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2

- Additive inverse: addition mod K yields 0.

- "Decrypt" by adding inverse.

$$1+9 \equiv 0 \pmod{10}$$

$$2+8 \equiv 0 \pmod{10}$$

$$9+1 \equiv 0 \pmod{10}$$

Encryption

- Plaintext = 'HELLO WORLD'
- $c = (a + m) \bmod 26$
- Let $a = 3$
- H: $(3 + 7) \bmod 26 = 10 \rightarrow K$
- E: $(3 + 4) \bmod 26 = 7 \rightarrow H$
- L: $(3 + 11) \bmod 26 = 14 \rightarrow O$
- L: $(3 + 11) \bmod 26 = 14 \rightarrow O$
- O: $(3 + 14) \bmod 26 = 17 \rightarrow R$
- (space)
- W: $(3 + 22) \bmod 26 = 25 \rightarrow Z$
- O: $(3 + 14) \bmod 26 = 17 \rightarrow R$
- R: $(3 + 17) \bmod 26 = 20 \rightarrow U$
- L: $(3 + 11) \bmod 26 = 14 \rightarrow O$
- D: $(3 + 3) \bmod 26 = 6 \rightarrow G$
- Ciphertext = 'KHOOR ZRUOG'

Decryption

- Plaintext = 'KHOOR ZRUOG'

- $m = (c - a) \bmod 26$

- Given $a = 3$, we can use the additive inverse as -3

- K: $(10 - 3) \bmod 26 = 7 \rightarrow H$

- H: $(7 - 3) \bmod 26 = 4 \rightarrow E$

- O: $(14 - 3) \bmod 26 = 11 \rightarrow L$

- O: $(14 - 3) \bmod 26 = 11 \rightarrow L$

- R: $(17 - 3) \bmod 26 = 14 \rightarrow O$

Actually, the additive inverse of 3 is 7. try it with adding 7 and then calculate

- (space)

- Z: $(25 - 3) \bmod 26 = 22 \rightarrow W$

- R: $(17 - 3) \bmod 26 = 14 \rightarrow O$

- U: $(20 - 3) \bmod 26 = 17 \rightarrow R$

- O: $(14 - 3) \bmod 26 = 11 \rightarrow L$

- G: $(6 - 3) \bmod 26 = 3 \rightarrow D$

- Ciphertext = 'HELLO WORLD'

Modular Multiplication

$(a \times b) \bmod 10$

- Multiplication modulo K

*	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7

- Multiplicative inverse: multiplication mod K yields 1
- Only some numbers have inverse (co-prime)

$$(3 \times 7) \bmod 10 = 1$$

Modular Multiplication

- Only the numbers *relatively prime* to n will have mod n multiplicative inverse
- x, m relative prime: no other common factor than 1
 - Eg. 8 & 15 are relatively prime - factors of 8 are 1,2,4,8 and of 15 are 1,3,5,15 and 1 is the only common factor.
 - In other words $\gcd(8, 15) = 1$

Example

- Alphabets

A=0 B=1 C=2 D=3 E=4 F=5 G=6 H=7
I=8 J=9 K=10 L=11 M=12 N=13 O=14 P=15
Q=16 R=17 S=18 T=19 U=20 V=21 W=22 X=23
Y=24 Z=25

- Module $K = 26$

- Key $a = 5$ (to satisfy $\gcd(a, 26) = 1$)

- Encryption: $c = (a \times m) \bmod 26$

- Multiplicative Inverse of $a \bmod 26$ is

$$a^{-1} = 21 \quad \text{as } (5 \times 21) = 105 = 1 \pmod{26}$$

- Decryption: $m = (a^{-1} \times c) \bmod 26$

Encryption

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7
I=8	J=9	K=10	L=11	M=12	N=13	O=14	P=15
Q=16	R=17	S=18	T=19	U=20	V=21	W=22	X=23
Y=24	Z=25						

- Plaintext = 'HELLO WORLD'
- $c = (a \times m) \bmod 26$
- Let $a = 5$, as $\gcd(5, 26) = 1$
- H: $(5 \times 7) \bmod 26 = 9 \rightarrow J$
- E: $(5 \times 4) \bmod 26 = 20 \rightarrow U$
- L: $(5 \times 11) \bmod 26 = 3 \rightarrow D$
- L: $(5 \times 11) \bmod 26 = 3 \rightarrow D$
- O: $(5 \times 14) \bmod 26 = 18 \rightarrow S$
- (space)
- W: $(5 \times 22) \bmod 26 = 6 \rightarrow G$
- O: $(5 \times 14) \bmod 26 = 18 \rightarrow S$
- R: $(5 \times 17) \bmod 26 = 9 \rightarrow H$
- L: $(5 \times 11) \bmod 26 = 3 \rightarrow D$
- D: $(5 \times 3) \bmod 26 = 15 \rightarrow P$
- Ciphertext = 'JUDDS GSHDP'

Decryption

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7
I=8	J=9	K=10	L=11	M=12	N=13	O=14	P=15
Q=16	R=17	S=18	T=19	U=20	V=21	W=22	X=23
Y=24	Z=25						

- Ciphertext = 'JUDDS GSHDP'
- $m = (a^{-1} \times c) \bmod 26$
- $a^{-1} = 21$ as
$$5 \times 21 = 105 = 1 \bmod 26$$
- J: $(21 \times 9) \bmod 26 = 7 \rightarrow H$
- U: $(21 \times 20) \bmod 26 = 4 \rightarrow E$
- D: $(21 \times 3) \bmod 26 = 11 \rightarrow L$
- D: $(21 \times 3) \bmod 26 = 11 \rightarrow L$
- S: $(21 \times 18) \bmod 26 = 14 \rightarrow O$
- (space)
- G: $(21 \times 6) \bmod 26 = 22 \rightarrow W$
- S: $(21 \times 18) \bmod 26 = 14 \rightarrow O$
- H: $(21 \times 7) \bmod 26 = 17 \rightarrow R$
- D: $(21 \times 3) \bmod 26 = 11 \rightarrow L$
- P: $(21 \times 15) \bmod 26 = 3 \rightarrow D$
- Plaintext = 'HELLO WORLD'

Totient Function

- Totient function $\phi(n)$: number of integers less than n relatively prime to n
 - if n is prime,
 - $\phi(n) = n - 1$
 - if $n = p \times q$, and p, q are primes, $p \neq q$
 - $\phi(n) = (p-1)(q-1)$
 - if $n = p_1 \times p_2 \times p_3 \times \dots \times p_m$, where p_i are prime factors
 - $\phi(n) = n \prod (1 - 1/p_i)$
 - E.g.,
 - $\phi(37) = 37 - 1 = 36$
 - $\phi(21) = (3-1) \times (7-1) = 2 \times 6 = 12$
 - $\phi(21) = 2^2 \times 5 = 20 (1-1/2) (1-1/5) = 8$

Multiplicative Inverse

- Is $a = 5$ and $a^{-1} = 21$ are the only numbers?
- Other combinations relatively prime to 26
- To ensure that multiplicative inverse exists
 $\gcd(a,b) = 1$
- How many co-primes for 26?
- $\phi(26) = (13-1) \times (2-1) = 12 \times 1 = 12$
- $\phi(26) = \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25\}$

Multiplicative Inverse

- $(a \times a^{-1}) \bmod 26 = 1$

Only numbers co-prime with 26 have multiplicative inverse

- 1: $(1 \times 1) \bmod 26 = 1$

- 2: $(3 \times 9) \bmod 26 = 1$

- 3: $(5 \times 21) \bmod 26 = 1$

- 4: $(7 \times 15) \bmod 26 = 1$

- 5: $(9 \times 3) \bmod 26 = 1$

- 6: $(11 \times 19) \bmod 26 = 1$

- 7: $(15 \times 7) \bmod 26 = 1$

- 8: $(17 \times 23) \bmod 26 = 1$

- 9: $(19 \times 11) \bmod 26 = 1$

- 10: $(21 \times 5) \bmod 26 = 1$

- 11: $(23 \times 17) \bmod 26 = 1$

- 12: $(25 \times 25) \bmod 26 = 1$

Quiz

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7
I=8	J=9	K=10	L=11	M=12	N=13	O=14	P=15
Q=16	R=17	S=18	T=19	U=20	V=21	W=22	X=23
Y=24	Z=25.	\$=26.	&=27.	%=28	#=29		

- Suppose you have the above set of alphabets
- Find
 - $\Phi(30) = ?$
- Choose a and a^{-1}
- Perform Encryption and Decryption
- P-text = "POWER\$#" (Note: POWER\$# is green in the original image)
- C-text = ?

Modular Exponentiation

$$c \equiv m^e \pmod{K}$$

xy	0	1	2	3	4	5	6	7	8	9
0		0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	1	2	4	8	6	2	4	8	6	2
3	1	3	9	7	1	3	9	7	1	3
4	1	4	6	4	6	4	6	4	6	4
5	1	5	5	5	5	5	5	5	5	5
6	1	6	6	6	6	6	6	6	6	6
7	1	7	9	3	1	7	9	3	1	7
8	1	8	4	2	6	8	4	2	6	8
9	1	9	1	9	1	9	1	9	1	9

Euler's Theorem

Let n be a positive integer, and x an integer such that:

$$\gcd(x, n) = 1$$

Then:

$$x^{\phi(n)} \equiv 1 \pmod{n}$$

Alternative Form (Useful for Exponents)

If y is any integer:

$$x^y \pmod{n} = x^{y \bmod \phi(n)} \pmod{n}$$

- This works when $\gcd(x, n) = 1$
- If $y \equiv 1 \pmod{\phi(n)}$, then:

$$x^y \equiv x \pmod{n}$$

- Encryption: $c = m^e \bmod K$,
where $e = \text{coprime } \phi(K)$
- $\phi(26) = 2 \times 13 = (2-1) \times (13-1) = 1 \times 12 = 12$
- Choose e that is co-prime to 12 [$\phi\{\phi(26)\} = 4$]
- Co-prime to $\phi(12) = \{1, 5, 7, 11\}$
 - In other words, $\gcd(e, \phi(26)) = \gcd(e, 12) = 1$
 - Let $e=5$, $\gcd(5,12) = 1 \rightarrow$ inverse exists $\rightarrow$ decrypt

Encryption

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7
I=8	J=9	K=10	L=11	M=12	N=13	O=14	P=15
Q=16	R=17	S=18	T=19	U=20	V=21	W=22	X=23
Y=24	Z=25						

- Plaintext = 'HELLO WORLD'
- $c = m^e \bmod 26$
- We have $e=5$
- H: $7^5 \bmod 26 = 11 \rightarrow L$
- E: $4^5 \bmod 26 = 10 \rightarrow K$
- L: $11^5 \bmod 26 = 7 \rightarrow H$
- L: $11^5 \bmod 26 = 7 \rightarrow H$
- O: $14^5 \bmod 26 = 14 \rightarrow O$
- (space)
- W: $22^5 \bmod 26 = 16 \rightarrow Q$
- O: $14^5 \bmod 26 = 14 \rightarrow O$
- R: $17^5 \bmod 26 = 23 \rightarrow X$
- L: $11^5 \bmod 26 = 7 \rightarrow H$
- D: $3^5 \bmod 26 = 9 \rightarrow J$
- Ciphertext = 'LKHHO QOXHJ'

- Decryption: $m = c^d \bmod K$ where
 - d when multiplied by e gives a remainder of 1 mod $\phi(K)$ i.e. $d \cdot e \equiv 1 \pmod{\phi(26)}$
 - d is the modular inverse of $e \bmod \phi(K)$
 - Only works if $\gcd(e, \phi(K)) = 1$
 - Compute d by trial and error (for small numbers)
 - For large numbers Extended Euclidean Algorithm,

- Given that $e=5$, solve for $5 \times d \equiv 1 \pmod{\phi(26)}$
 - For $d=1$? $5 \times 1 = 5 \not\equiv 1 \pmod{12}$
 - For $d=2$? $5 \times 2 = 10 \not\equiv 1 \pmod{12}$
 - For $d=3$? $5 \times 3 = 15 \not\equiv 1 \pmod{12}$
 - For $d=4$? $5 \times 4 = 20 \not\equiv 1 \pmod{12}$
 - For $d=5$? $5 \times 5 = 25 \equiv 1 \pmod{12}$
 - We can choose $d = 5$

Decryption

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7
I=8	J=9	K=10	L=11	M=12	N=13	O=14	P=15
Q=16	R=17	S=18	T=19	U=20	V=21	W=22	X=23
Y=24	Z=25						

- Ciphertext = 'LKHHO QOXHJ'
- $m = c^d \bmod 26$
as calculated $d = 5$
- L: $11^5 \bmod 26 = 7 \rightarrow H$
- K: $10^5 \bmod 26 = 4 \rightarrow E$
- H: $7^5 \bmod 26 = 11 \rightarrow L$
- H: $7^5 \bmod 26 = 11 \rightarrow L$
- O: $14^5 \bmod 26 = 14 \rightarrow O$
- (space)
- Q: $16^5 \bmod 26 = 22 \rightarrow W$
- O: $14^5 \bmod 26 = 14 \rightarrow O$
- X: $23^5 \bmod 26 = 17 \rightarrow R$
- H: $7^5 \bmod 26 = 11 \rightarrow L$
- J: $9^5 \bmod 26 = 3 \rightarrow D$
- Plaintext = 'HELLO WORLD'

Quiz

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7
I=8	J=9	K=10	L=11	M=12	N=13	O=14	P=15
Q=16	R=17	S=18	T=19	U=20	V=21	W=22	X=23
Y=24	Z=25.	\$=26.	&=27.	%=28	#=29		

- Encryption:

- $C = m^e \bmod K$
- $K? e?$

- Decryption:

- $M = c^d \bmod K$
- $d?$

- P-text =
"POWER\$#"

- C-text = ?

Comparison

Plaintext = 'HELLO WORLD'

Method	Formula	Paramter	Ciphertext
Modular Addition	$c = (m + a) \bmod 26$	$a = 3$	LKHHO QOXHJ
	$m = (c + (-a)) \bmod 26$	$a = -3 \text{ or } 7$	
Modular Multiplication	$c = (m \times a) \bmod 26$	$a = 5$	KHOOR ZRUOG
	$m = (c \times a^{-1}) \bmod 26$	$a^{-1} = 21$	
Modular Exponentiation	$c = m^e \bmod 26$	$e = 5$	JUDDS GSHDP
	$m = c^d \bmod 26$	$d = 5$	

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- Modular Arithmetic
- RSA

RSA (Rivest, Shamir, Adleman)

- The most popular one.
- Support both public key encryption and digital signature.
- Assumption/theoretical basis:
 - Factoring a big number is hard.
- Variable key length (usually 512 bits).
- Variable plaintext block size.
 - Plaintext must be "smaller" than the key.
 - Ciphertext block size is the same as the key length.

What Is RSA?

- To generate key pair:
 - Pick large primes (≥ 256 bits each) p and q
 - Let $n = p * q$, where n is public, but p and q are private
 - For public key,
 - choose e that is relatively prime to $\phi(n) = (p-1)(q-1)$,
 - let public Key = $\langle e, n \rangle$
 - For private key
 - find d that is the multiplicative inverse of $e \bmod \phi(n)$, i.e., $e * d = 1 \bmod \phi(n)$,
 - let private key = $\langle d, n \rangle$

RSA Example

Select primes: $p=17$ & $q=11$

Compute $n = pq = 17 \times 11 = 187$

Compute $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$

Select e : $\gcd(e, 160) = 1$; choose $e=7$

Determine d : $d \times e \equiv 1 \pmod{160}$ and $d < 160$

Value is $d=23$

since $23 \times 7 = 161 = 10 \times 160 + 1$ or $161 \equiv 1 \pmod{160}$

Publish public key $KU = \{7, 187\} = \{e, n\}$

Keep secret private key $KR = \{23, 17, 11\} = \{d, p, q\}$

How Does RSA Work?

- Given $\text{pub} = \langle e, n \rangle$ and $\text{priv} = \langle d, n \rangle$
 - encryption: $c = m^e \bmod n, m < n$
 - decryption: $m = c^d \bmod n$
 - signature: $s = m^d \bmod n, m < n$
 - verification: $m = s^e \bmod n$
- given message $M = 88$ (nb. $88 < 187$)
- encryption:
$$C = 88^7 \bmod 187 = 11$$
- decryption:
$$M = 11^{23} \bmod 187 = 88$$

Why Does RSA Work?

- Given $\text{pub} = \langle e, n \rangle$ and $\text{priv} = \langle d, n \rangle$
 - $n = p \cdot q, \phi(n) = (p-1)(q-1)$
 - $e \cdot d = 1 \bmod \phi(n)$
 - $x^{e \cdot d} = x \bmod n$
 - encryption: $c = m^e \bmod n$
 - *[anyone with e and n can encrypt $\rightarrow$ public key]*
 - decryption: $m = c^d \bmod n$
 - *[Only owner of private key $\langle d, n \rangle$ can decrypt]*
 - *Euler's theorem: $c^d = (m^e)^d = m^{1 + k \phi(n)} \equiv m (m^{\phi(n)})^k \equiv m \cdot 1^k \equiv m \bmod n$
(works for $m < n$)*
 - digital signature (similar)

Summary

Operation	Formula	Key Used	Notes
Encryption	$c = m^e \bmod n$	Public e, n	Anyone can encrypt
Decryption	$m = c^d \bmod n$	Private d, n	Only owner can decrypt
Signature	$s = m^d \bmod n$	Private d, n	Sign with private key
Verify Signature	$m = s^e \bmod n$	Public e, n	Anyone can verify

Security of RSA

- A key of size 512 bits means
 - $2^{512} = 1.34 \times 10^{154}$
- Naïve Trial Division:
 - Check all integers $\leq \sqrt{2^{512}} = 2^{256}$
- Best Known Algorithm:
 - Number Field Sieve (sub-exponential, still extremely slow for 512bit)
 - RSA-155 (512-bit) took 8,400 MIPS-Years

Brute-force calculation

Key Size	Approx. Keyspace	Time to brute-force (seconds)	Time in years
512 bits	$2^{512} \approx 1.34 \times 10^{154}$	1.03×10^{142} sec	3.27×10^{134} years
1024 bits	$2^{1024} \approx 1.80 \times 10^{308}$	1.38×10^{296} sec	4.37×10^{288} years
2048 bits	$2^{2048} \approx 3.23 \times 10^{616}$	2.48×10^{604} sec	7.87×10^{595} years

Note: 1.3 TFLOPS = 1.3×10^{12} instructions/sec. Calculations assume **1 key test = 1 FLOP** (idealized). Real brute-force would be even slower.

Factorization is hard!

- Public key: n, e
- Private key: d (depends on $\varphi(n) = (p-1)(q-1)$)
- Security depends on factorization being hard
- Larger key $\rightarrow$ exponentially harder to factor
 - 512-bit: doable with effort
 - 1024-bit, 2048-bit: infeasible today

Is RSA Secure?

- Factoring 512-bit number is very hard!
- But if you can factor big number n then given public key $\langle e, n \rangle$, you can find d , hence the private key by:
 - Knowing factors p, q , such that, $n = p * q$
 - Then $\phi(n) = (p-1)(q-1)$
 - Then d such that $e * d = 1 \bmod \phi(n)$
- Threat
 - Moore's law - faster CPU, Parallelism
 - Refinement of factorizing algorithms
- For the near future, a key of 1024 or 2048 bits needed

Symmetric (DES) vs. Public Key (RSA)

- Exponentiation of RSA is expensive !
- AES and DES are much faster
 - 100 times faster in software
 - 1,000 to 10,000 times faster in hardware
- RSA often used in combination in AES and DES
 - Pass the *session key* with RSA